MASHUK ELAHI

Marine & Offshore Electrical Engineer | Hybrid Power, Vessel Electrification & Engineering Digitalisation

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PROFESSIONAL SUMMARY

Marine electrical engineer with 16 years delivering LV/MV/HV power systems for vessel newbuilds, offshore assets and offshore-wind applications, now focused on hybrid-electric vessel power. Specialises in class-compliant electrical design, MV 11 kV protection, SLDs and cable schedules, BESS and power-management-system integration, and commissioning through handover under ABS, BV, LR and DNV. Currently delivering a hybrid-power architecture for a 42 m vessel newbuild, cutting fuel draw by up to 50% at low load, and running the engineering digitalisation toolchain that generates its cable schedule, load list, alarm point list and drawing register deterministically from one data set.

CORE COMPETENCIES

- LV/MV/HV Power Distribution & Class-Compliant Design
- MV 11 kV Protection & IEC 61850 Systems
- BESS, BMS & Hybrid Propulsion Integration
- Power-Management Systems & Shaft Generators
- SLDs, Load Lists & Cable Schedules
- Engineering Digitalisation & Design Automation
- ABS, BV, LR & DNV Class Approvals
- IEC 60092, IEC 61892, IEC 60364 & SOLAS
- ABB 800xA, FAT, SAT & FMEA
- Energisation, Sea Trials, Sail-Away & Handover
- Newbuild, FPSO, Jack-Up Rig & Offshore-Wind Substation Work

PROFESSIONAL EXPERIENCE

Electrical Engineer | Hybrid Marine & Vessel Electrification

Strategic Marine Pte. Ltd., Singapore | Jul 2024 - Present

- Designed LV power distribution for a multi-generator system with three prime-power generator sets totalling 450 ekW; delivered single-line diagrams, load lists and cable schedules, securing first-pass approval from four independent certification bodies with zero rework.
- Integrated a 72 kWh lithium-ion BESS, battery management system and DC/DC inverters into a hybrid power architecture, reducing fuel draw by up to 50% during low-load operation.
- Led the BMS-to-power-management-system interface work and Factory Acceptance Testing through to class-compliant handover.
- Engineered aluminium-hull IT earthing architecture with single-point common-zero bonding to IEC 60092-201, and cleared stray-current loops on fast crew boats by upgrading isolation switches to DPST.
- Built the project's engineering digitalisation toolchain: deterministic Python generation of the cable schedule, cable order summary, electrical load list, AMS alarm point list and drawing register from one data set, with a compliance linter as the issue gate and byte-identical rebuilds as the release test.
- Wrote the yard's own drawing application in Python and Qt — its own drawing format, a 610-symbol library built from the project's drawings, SVG / DXF / PDF output, a 15-check verifier, and a tool interface through which an AI agent authors and marks up sheets.
- Tracked electrical-system weights exceeding 7,800 kg across load lists and cable schedules, feeding vessel stability calculations.

Senior Project Engineer | Critical-Power Systems & Prefabrication

Exyte, Singapore | Feb 2023 - Jul 2024

- Led 300 mm semiconductor cleanroom critical-power hook-up, energisation and Functional Performance Testing on 30 MW (33 MVA) of critical load at 1.5 kW/m² fab power density, achieving zero audit rework and first-pass PLC/SCADA interlock validation.
- Introduced prefabricated electrical power skids, reducing on-site installation labour by 50% and removing contamination risk during live energisation — directly transferable to modular vessel and offshore power builds.

E&I Engineer | Renewable, Cogeneration & Hazardous Areas

Phoenix Mecano S.E.A. Pte. Ltd., Singapore | Jan 2021 - Jan 2023

- Delivered Malaysia's first solar-wind autonomous remote-power installation using a 24 V DC solar-battery bus, eliminating routine diesel-generator use and enabling remote/unmanned operation.
- Directed the E&I scope for an onshore 10 MW cogeneration plant — transformers, MV/LV switchgear, a 400 kVA UPS, SCADA integration, tie-in design and instrument loops — and designed IECEx/ATEX hazardous-area systems including Ex equipment selection, intrinsically safe barriers and zoning studies.

E&I Superintendent | Offshore Power Systems & Commissioning

Keppel FELS / Keppel Offshore & Marine, Singapore | Nov 2012 - Dec 2020

- Planned and monitored the electrical cabling programme for a jack-up rig newbuild — a scope estimated at 430,000 m of cable and 365,500 manhours — tracking weekly cabling metres and electrical-connection points by zone and phase against a 0.85 manhour-per-metre productivity target, including subcontractor loading.
- Delivered medium-voltage thruster power distribution on semi-submersible newbuilds: thruster transformers feeding MV7000 MV converters across two thruster rooms, with cable routing and termination through lower hull, forward columns and living quarters.
- Produced the E&I drawing and as-built package for an FPSO topsides scope (MODEC): 11 kV switchgear connection diagrams, 440 V essential and process switchgear, 230 V lighting and UPS boards, fire-and-gas and PAGA block and connection diagrams across four topside modules, module earthing schematics, instrument junction-box and cable block diagrams, and hazardous-area LED lighting selection with emergency battery backup.
- Led commissioning of MV 11 kV switchgear and ABB 800xA SCADA-based fire-and-gas systems, maintaining MV/LV power reliability through final handover with crews of up to 100 personnel.
- Energised an offshore-wind substation electrical system to full IEC 61850 protection-relay conformity, achieving zero punch-list items at handover.
- Produced the electrical schematics for an autonomous tug — ABB/Niigata propulsion interface, vessel-care system and a 240 V power panel for autonomous operation — and delivered the electrical scope of a 12 MW, 3,750 m³ floating living-lab vessel.
- Ran cable termination campaigns with subcontract crews against a daily tags-per-man plan; authored 30 E&I risk assessments and mentored 15 engineers through Innovation Quality Circles.

Assistant Electrical Engineer | Vessel Newbuild Electrical

Ananda Shipyard & Slipways Ltd., Bangladesh | Sep 2009 - Oct 2012

- Supported electrical installation, testing and commissioning activities on vessel newbuild projects, assisting senior engineers with equipment energisation and documentation.

EDUCATION

- Master's Degree in Continuing Education in Sustainable Electricity Generation (UCAM, 60 ECTS, título propio) — UCAM Universidad Católica de Murcia, Spain — Apr 2026. Dissertation: Integration of Short and Long Duration Energy Storage Systems with Renewable Energy and EV Charging across European Countries (sole-authored systematic review).
- MSc Digital Transformation (Pass with Merit) — London Metropolitan University, UK — Apr 2026. Work Based Project: A Digital Transformation Strategy for Integrated Commissioning, Telemetry, and Inventory Control at Strategic Marine Pte. Ltd.
- Master of Business Administration (MBA) — University of Sunderland, UK — Jun 2025
- BSc Electrical and Electronic Engineering — Bangladesh University — Jul 2008. Final-year project: design and construction of an automatic transfer switch with a backup control system for a hybrid power plant.
- Diploma in Electronics Engineering — Jessore Polytechnic Institute — Jan 2004

CERTIFICATIONS

- Certified Data Centre Professional (CDCP), EPI — 2026
- ISO 50001:2018 Energy Management Lead Auditor, GAQM / IGC
- ISO 9001:2015 Quality Management Lead Auditor, GAQM / IGC
- IECEx Ex 001: Basic Principles of Protection in Explosive Atmospheres — valid through Oct 2027
- ABB Power System / LV Switchboard Training
- Fluke DSX2-8000 Cable Analyzer / Copper Certifier Training

TECHNICAL TOOLS

Python (deterministic document generation); AutoCAD Electrical; DXF and PDF automation; ABB 800xA; PLC/SCADA systems; IEC 61850 protection relays; EPMS/SCADA; Fluke DSX2-8000 Cable Analyzer / Copper Certifier.

PROFESSIONAL DEVELOPMENT

Areas of active study and continuing development (not claimed as project-verified experience):

- IACS UR E26 and E27 (vessel cyber-resilience) - in progress
- IEC 62443 industrial / marine cybersecurity - professional development
- Shore power / cold-ironing systems - professional development
- DC / LVDC zonal distribution architectures - professional development
- IEC 61660 (DC short-circuit) and IEC 62619 (industrial Li-ion safety) - in progress